

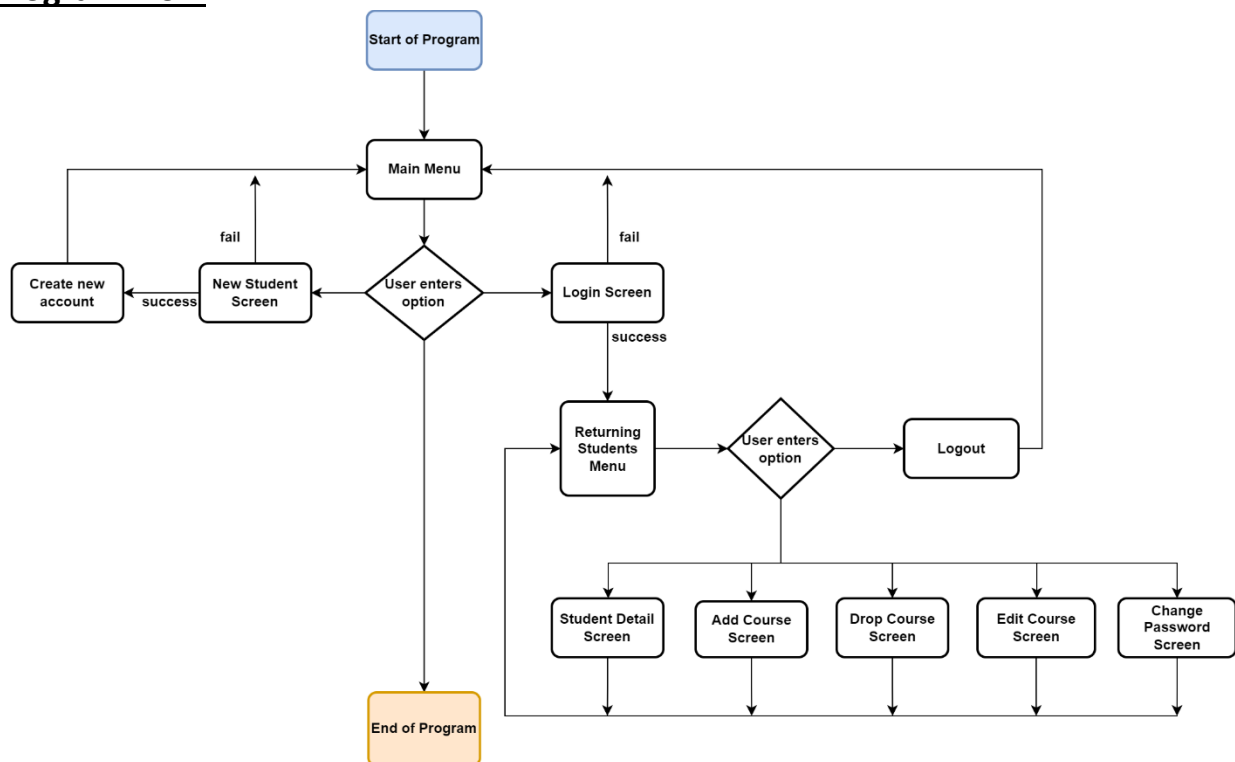
Assignment 2 –Student Management System

1. General Info

You are to help **CIMP** to write Student Management System (**StuSys**). The system will be used by guidance to help keep track of student's marks and other information. The system has the following requirements:

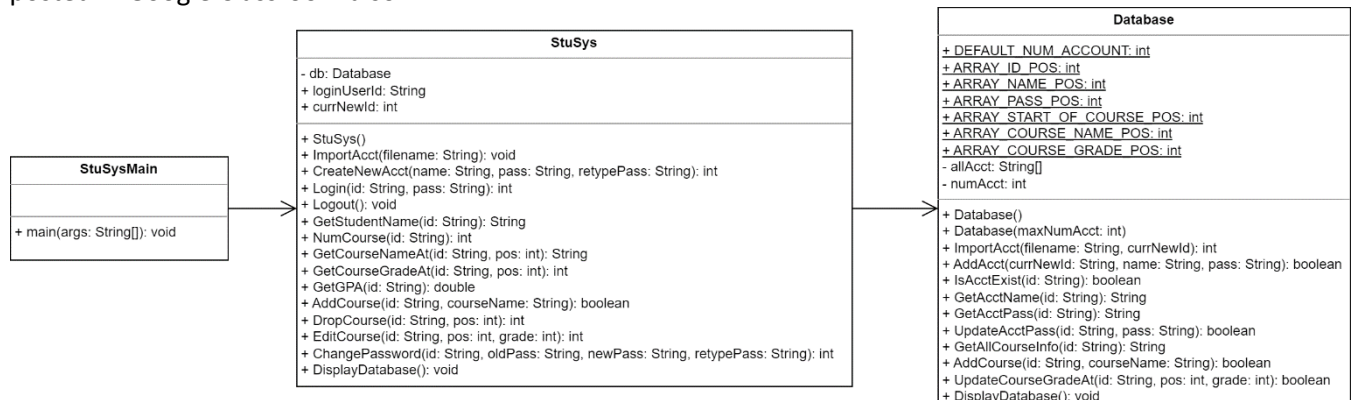
- Has a login system, where each student has a login password.
- Has a create new student function, for new students to be entered into the system.
- Once logged into a student's account, she/he has the following options:
 - Student detail info
 - Add a course
 - Drop a course
 - Edit a course grade
 - Change password
 - Logout

2. Program Flow



3. UML Diagram

Here's the UML diagram of the program. Your program must follow the same structure. There's a larger version posted in Google Classroom also.



4. User Interface

a) Main Screen

```
=====
|  WELCOME TO CIMP STUDENT SYSTEM  |
=====
1. New Student
2. Returning Student
3. Exit System
Pick an option:
```

b) New Student Screen

```
=====
|  WELCOME TO CIMP STUDENT SYSTEM  |
=====
1. New Student
2. Returning Student
3. Exit System
Pick an option: 1

-----
|      CREATE ACCOUNT      |
-----
Student ID      : 10004
Full Name       : KW
Password        : 12345
Retype Password : 12345
Successfully created account '10004'
```

c) “Login Screen” (For Returning Students) & “Returning Student Menu”

```
=====
|  WELCOME TO CIMP STUDENT SYSTEM  |
=====
1. New Student
2. Returning Student
3. Exit System
Pick an option: 2

-----
|      LOGIN SCREEN      |
-----
Student ID : 10004
Password   : 12345
Login Success.

=====
|      CIMP STUDENT SYSTEM      |
|      OPTION SCREEN           |
|=====|
| 1. STUDENT DETAIL | 4. EDIT COURSE GRADE |
|=====|
| 2. ADD COURSES    | 5. CHANGE PASSWORD  |
|=====|
| 3. DROP COURSES   | 6. LOGOUT           |
|=====|
Pick an option:
```

d) Student Detail Screen (student without any courses)

```

=====
|                                |
|          CIMP STUDENT SYSTEM  |
|          OPTION SCREEN        |
|                                |
|-----|-----|
| 1. STUDENT DETAIL | 4. EDIT COURSE GRADE |
|-----|-----|
| 2. ADD COURSES   | 5. CHANGE PASSWORD  |
|-----|-----|
| 3. DROP COURSES  | 6. LOGOUT            |
|-----|-----|
|                                |
| Pick an option:          |
| 1                         |
|                                |
|-----|-----|
|          STUDENT DETAIL      |
|-----|-----|
| Student ID   :          10004 |
| Student name :          KW   |
| GPA          :          N/A   |
|-----|-----|
|                                |
|-----|-----|
|          COURSES   |          Grades   |
|-----|-----|
|                                |
| No course at the moment! |
|                                |
=====

```

Based on course taken. No course right now, so displays N/A

e) Student Detail Screen (student with completed courses)

```

=====
|                                |
|          CIMP STUDENT SYSTEM  |
|          OPTION SCREEN        |
|                                |
|-----|-----|
| 1. STUDENT DETAIL | 4. EDIT COURSE GRADE |
|-----|-----|
| 2. ADD COURSES   | 5. CHANGE PASSWORD  |
|-----|-----|
| 3. DROP COURSES  | 6. LOGOUT            |
|-----|-----|
|                                |
| Pick an option:          |
| 1                         |
|                                |
|-----|-----|
|          STUDENT DETAIL      |
|-----|-----|
| Student ID   :          10001 |
| Student name :        Sunny Wong |
| GPA          :        76.33 %   |
|-----|-----|
|                                |
|-----|-----|
|          COURSES   |          Grades   |
|-----|-----|
| 1 MHF4U          |          75   |
| 2 ENG4U          |          71   |
| 3 AMU4M          |          83   |
| 4 MDM4U          |          TBD   |
|-----|-----|
|                                |
=====

```

This student has has courses. So it will calculate the GPA (average) of the courses completed only. Do not include TBD or Dropped courses.

f) Add Course Screen

```
=====
|                                |
|          CIMP STUDENT SYSTEM  |
|          OPTION SCREEN        |
|=====|=====|
| 1. STUDENT DETAIL | 4. EDIT COURSE GRADE |
|-----|-----|
| 2. ADD COURSES    | 5. CHANGE PASSWORD  |
|-----|-----|
| 3. DROP COURSES   | 6. LOGOUT            |
|=====|=====|
Pick an option:
2

=====
|          ADD COURSES          |
|=====|
Course to add: ICS4U
Successfully added 'ICS4U'
```

After adding the course, the “Student Detail” screen will reflect that.

```
=====
|                                |
|          CIMP STUDENT SYSTEM  |
|          OPTION SCREEN        |
|=====|=====|
| 1. STUDENT DETAIL | 4. EDIT COURSE GRADE |
|-----|-----|
| 2. ADD COURSES    | 5. CHANGE PASSWORD  |
|-----|-----|
| 3. DROP COURSES   | 6. LOGOUT            |
|=====|=====|
Pick an option:
1

=====
|          STUDENT DETAIL        |
|=====|=====|
| Student ID   :      10001      |
| Student name :    Sunny Wong   |
| GPA          :      76.33 %    |
|=====|=====|

=====
| COURSES | Grades |
|-----|-----|
| 1 MHF4U |      75 |
| 2 ENG4U |      71 |
| 3 AMU4M |      83 |
| 4 MDM4U |     TBD |
| 5 ICS4U |     TBD |
|=====|=====|
```

After adding a course, it will show here as TBD (To Be Determined or not completed yet)

g) Drop Course Screen

```

=====
CIMP STUDENT SYSTEM
OPTION SCREEN
=====
1. STUDENT DETAIL      | 4. EDIT COURSE GRADE
-----|-----
2. ADD COURSES         | 5. CHANGE PASSWORD
-----|-----
3. DROP COURSES        | 6. LOGOUT
=====

Pick an option:
3

=====
DROP COURSES
=====
1 MHF4U      | 75
2 ENG4U      | 71
3 AMU4M      | 83
4 MDM4U      | TBD
5 ICS4U      | TBD
=====

Select course number to drop: 2
Successfully dropped the course

```

After dropping the course, the “Student Detail” screen will reflect that.

```

=====
CIMP STUDENT SYSTEM
OPTION SCREEN
=====
1. STUDENT DETAIL      | 4. EDIT COURSE GRADE
-----|-----
2. ADD COURSES         | 5. CHANGE PASSWORD
-----|-----
3. DROP COURSES        | 6. LOGOUT
=====

Pick an option:
1

=====
STUDENT DETAIL
=====
Student ID   : 10001
Student name : Sunny Wong
GPA          : 79.00 %
=====

=====
COURSES      | Grades
=====
1 MHF4U      | 75
2 ENG4U      | DROPPED
3 AMU4M      | 83
4 MDM4U      | TBD
5 ICS4U      | TBD
=====

```

ENG4U is dropped. So it will show it here. Also GPA will ignore this course.

h) Edit Course Screen

```
=====
CIMP STUDENT SYSTEM
OPTION SCREEN
=====
1. STUDENT DETAIL      | 4. EDIT COURSE GRADE
-----|-----
2. ADD COURSES         | 5. CHANGE PASSWORD
-----|-----
3. DROP COURSES        | 6. LOGOUT
=====

Pick an option:
4

=====
EDIT COURSE GRADE
=====
YOUR CURRENT COURSES:
1 MHF4U      | 75
2 ENG4U      | DROPPED
3 AMU4M      | 83
4 MDM4U      | TBD
5 ICS4U      | TBD
=====

Select course number to edit: 4
Enter the new grade: 99
Successfully edited the course.
```

After editing the course, the “Student Detail” screen will reflect that.

```
=====
CIMP STUDENT SYSTEM
OPTION SCREEN
=====
1. STUDENT DETAIL      | 4. EDIT COURSE GRADE
-----|-----
2. ADD COURSES         | 5. CHANGE PASSWORD
-----|-----
3. DROP COURSES        | 6. LOGOUT
=====

Pick an option:
1

=====
STUDENT DETAIL
=====
Student ID   : 10001
Student name : Sunny Wong
GPA          : 85.67 %
=====

=====
COURSES      | Grades
=====
1 MHF4U      | 75
2 ENG4U      | DROPPED
3 AMU4M      | 83
4 MDM4U      | 99
5 ICS4U      | TBD
=====
```

MDM4U was edited and it shows the new grade. The GPA also reflects the new average

I) Change Password Screen

```

=====
|                                |
|          CIMP STUDENT SYSTEM  |
|          OPTION SCREEN        |
|                                |
|=====|=====|
| 1. STUDENT DETAIL | 4. EDIT COURSE GRADE |
|-----|-----|
| 2. ADD COURSES   | 5. CHANGE PASSWORD  |
|-----|-----|
| 3. DROP COURSES  | 6. LOGOUT                |
|=====|=====|
Pick an option:
5

|-----|
|          CHANGE PASSWORD      |
|-----|
Enter old password : 11111
Enter new password : 12345
Retype Password   : 12345
Successfully changed the account password

```

j) Logout from the “Returning Student Menu”

```

=====
|                                |
|          CIMP STUDENT SYSTEM  |
|          OPTION SCREEN        |
|                                |
|=====|=====|
| 1. STUDENT DETAIL | 4. EDIT COURSE GRADE |
|-----|-----|
| 2. ADD COURSES   | 5. CHANGE PASSWORD  |
|-----|-----|
| 3. DROP COURSES  | 6. LOGOUT                |
|=====|=====|
Pick an option:
6

=====
| WELCOME TO CIMP STUDENT SYSTEM |
=====
1. New Student
2. Returning Student
3. Exit System
Pick an option:

```

k) Exit the system.

```

=====
| WELCOME TO CIMP STUDENT SYSTEM |
=====
1. New Student
2. Returning Student
3. Exit System
Pick an option: 3

System is shutting down.

```

5. How Info is Stored in the Database & Importing Text File

The **StuSys** allows you to initially import dummy data to be used for testing from a text file (.txt). This textfile is NOT the place where data are stored. So once the program exits, all of its memory (RAM) will be gone. This part of the program is already developed for you. You just need to learn how to import it into your Database class.

Text filename must be **AcctInfo.txt** and it must be in the same folder as the code for importing to work.

In the text file, the columns are separated by colons ":" and has the following format:

Column #	Item	Special Notes
1	Account Name	
2	Account Password	Must be 5 characters/digits
3 and onwards	Course Info	Each course contains 2 info, Course Code and Grade. They are separated by an underscore "_" Example: CourseCode1_Grade1

The database stores the accounts in a String array. Each block in the array contains only one student and it has the following format:

```
[Account ID]:[Student Name]:[Password]:[CourseCode1_Grade1]:[CourseCode2_Grade2]: ... :[CourseCodeN_GradeN]
```

Note:

- Each item is separated by a colon ":"
- Account ID starts at 10000, and it increments by 1 after it has been assigned
 - Ex: 10000, 10001, 10002...
- Password must contain exactly 5 characters/digits(No symbols)
- Students may and may not have a course.
- Grades must be integer value (no decimals)
- A course contains 2 information in the format **[CourseCode1_Grade1]**
 - **CourseCode** and **Grade** is separated by an underscore "_"
- If the grade is **-100**, it means the course is still in progress (The program should show **TBD** instead of **-100** on the screen)
- If the grade is **-200**, it means the course has been dropped (The program should show **DROPPED** instead of **-200** on the screen)

Extra – For those who wants to understand the code inside **Database.java** (NOT REQUIRED):

For more information on how to work with .txt using Java, please refer to the following websites:

- <https://www.geeksforgeeks.org/different-ways-reading-text-file-java/>
- <https://www.javatpoint.com/java-bufferedwriter-class>

6. Requirements

- Do not change the **Database** class. If Database is changed and program cannot be compiled, then you will get zero.
- Must use the **Database** class to help you store and access the data. Read the JAVA DOCS (Comments) to help you understand how to use each method inside the **Database** class. Just understand the method header is enough. It is not necessary to understand what is happening inside each of the method.
- Program must follow the program flow chart and UML diagram.
- Do not create extra classes
- Do not create extra public methods (except helper methods in **StuSys** class)
- Must complete all template methods and the main methods.
- Each class must be in its separate file to improve readability and maintainability.
- Do not use anything that's not taught in Unit 1 and Unit 2 (*Eg: ArrayList, LinkedList, 2D-Array, etc.*)
- Must create extra private methods (Part of the marking scheme)
- Must following Front-End and Back-End design.

7. Error Checking

- Program must check if password has 5 characters/digits, and warn the user accordingly if it's not.
- Must check if the new password and retyped password match.
- Must make sure users entered valid options (In Main screen, Logged In screen, Delete Course screen & Edit Course screen)
- Must check grade values are valid.
- Assume options entered will always be the correct data type (**Eg:** If the Scanner is expecting an int, I will not enter a string)

8. Provided Templates and Files to Submit

Provided Code & Template	Files to submit	Do not submit the following
● Database.java ● StuSys.java ● StuSysMain.java ● AcctInfo.txt	● StuSysMain.java ● StuSys.java	● Database.java ● AcctInfo.txt ● Any .class or .java~

9. Note

- You may write as many or as little code. However, try to aim for my benchmark.
- My sample code contains:
 - **303 lines** of code in StuSys.java
 - **302 lines** of code in StuSysMain.java
 - **605 lines** in total
- The templates contain a total of **161** lines of code already.
- You just need to write **444** lines of code to complete this assignment.
- Working with friends is highly encouraged, and may share ideas. However no sharing of code. There are programs that can check the program's similarity, and I can see who are copying programs.

10. Marking Scheme: Checklist

Knowledge		
Items	Your Mark	Out of
● Created and Used Methods correctly		4
● Used Classes and Objects correctly		4
● Display info correctly		4
● Program loops correctly between Menu and Screens		4
● Used material not from lesson		-8
Knowledge Total:		16
Application		
Items	Your Mark	Out of
● Constructor <code>StuSys()</code> is working		2
● <code>CreateNewAcct()</code> is working		4
● <code>Login()</code> is working		4
● <code>Logout()</code> is working		1
● <code>GetStudentName()</code> is working		2
● <code>NumCourse()</code> is working		2
● <code>GetCourseNameAt()</code> is working		4
● <code>GetCourseGradeAt()</code> is working		4
● <code>GetGPA()</code> is working		4
● <code>AddCourse()</code> is working		3
● <code>DropCourse()</code> is working		4
● <code>EditCourse()</code> is working		5
● <code>ChangePassword()</code> is working		6
● Used material not from lesson		-22
Application Total:		45
Thinking		
Items	Your Mark	Out of
● Created new Global Constant variables and used them correctly/efficiently throughout the program (minimum 5)		4
● Created Helper Methods and used them correctly/efficiently (minimum 2)		4
● Checking for invalid input		4
● Followed Front-End Back-End design		4
● Followed UML diagram design		8
● Used material not from lesson		-12
Thinking Total:		24
Communication		
Items	Your Mark	Out of
● Correct variable naming convention		4
● Sufficient comments in code		4
● Code indentation and organization		4
● Clear user interface with correct formatting (Based on Sample Images)		4
● Appropriate descriptive warning messages		4
Communication Total:		20
PERCENTAGE:		
Comments		